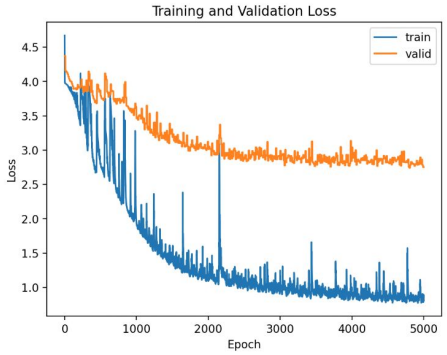
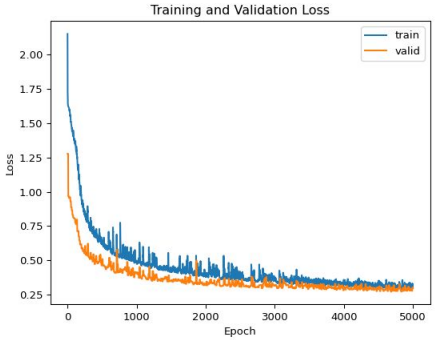
Mesh GNNs (MeshGraphNet) Loss function

1st Loss history



2nd Loss history



$$\mathcal{L} = \sqrt{\frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\tilde{y}_i - \hat{y}_i\|_2^2}$$

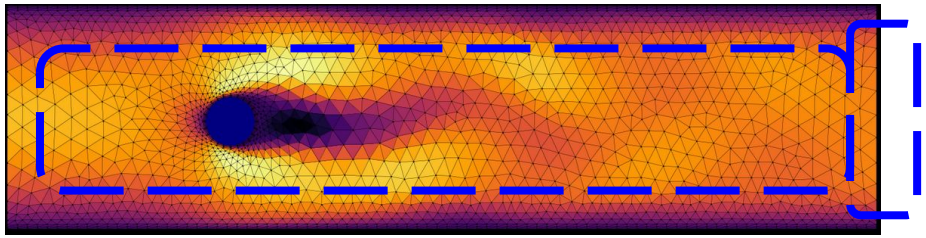
$\mathcal{M}$ is the set of nodes used in the Loss

$\tilde{y}_i$ is the normalized true target at node i

$\hat{y}_i$ is the predicted target at node i

```python
def loss(self, pred, inputs, mean_vec_y, std_vec_y):

    # Define  the node types we calculate the loss for.
    # Since our model has 4 node types (inlet, outlet, wall, internal)
    # and the data is was simulated from a prescribed inlet velocity and no-slip
    # boundary conditions at the walls, we only calculate the loss for the internal
    # and oulet nodes, as the inlet and wall nodes have prescribed dynamics.

    # normal means internal fluid nodes.

    normal = torch.tensor(0)
    outflow = torch.tensor(5)

    # Get the loss mask for the normal and outflow nodes
    loss_mask = torch.logical_or((torch.argmax(inputs.x[:, 2:], dim=1) == normal),
                                 (torch.argmax(inputs.x[:, 2:], dim=1) == outflow))

    # Normalize true values with dataset mean and std
    labels = norm.normalize(inputs.y, mean_vec_y, std_vec_y)

    # Sum of square errors for each node (in here considering all the nodes))
    error = torch.sum((labels - pred) ** 2, axis=1) # compute the error for each node, error is [num_nodes]

    # Root mean squared error for the nodes we are interested in (normal and outflow nodes)
    loss = torch.sqrt(torch.mean(error[loss_mask]))

    return loss
```

The loss seems reasonable considering that size of my dataset (coarse). I plan to add more time-steps to the training phase.



$\mathcal{M} = \text{normal+outflow}$

The biggest challenge so far was identifying why the loss was that high as in the first pass, this is due a mistaken I made in implementing the normalization functionality. There I was simply redefining quantities as mean_vec_x, not summing the normalization for each term (node and edge) in the graph.

As the next steps I plan to implement wandb log, rollout capabilities, deploy this in the university cluster to increase my training data.